

Asktrix CRM Mobile Agent App (APK)

Developer Requirement Document – Version 1.0

1. Objective

Develop a secure Android employee operations app integrated with the CRM. It must never expose customer information, prevent theft of customer data, allow only CRM-controlled communication, synchronize every activity with the CRM, secure company-owned devices, and provide complete management visibility.

2. Users

Customer Support, Sales Team, Documentation Team, Accounts Team, Relationship Managers, Team Leaders.

3. CRM Integration

Fetch all client information directly from CRM APIs. No permanent customer database inside the mobile. Use only temporary encrypted cache. Data includes Client Name, Client ID, Service ID, Assigned Employee, Process Status, Documents Pending, Payment Status, Government Status, Internal Remarks, Uploaded KYC, Timeline, Previous Calls, and Email History.

4. Client Information Visibility

Mask customer mobile numbers (98XXXXXX12) and email addresses (siv****@gmail.com). Employees must never see the full contact details.

5. Click-to-Call

Calls are initiated only through CRM. No dial pad, no number copying, and no visible phone number.

6. Call Recording

Automatically record, encrypt, upload to CRM, then delete locally after successful upload. Offline recordings remain encrypted until synced.

7. Call Log Sync

Synchronize incoming, outgoing, missed calls, duration, timestamps, employee ID and client ID.

8. CRM Timeline

Automatically update every interaction including calls, recordings, remarks, follow-ups, payments and next actions.

9. Internet Behaviour

Automatically detect connectivity and sync recordings, logs, assignments and updates without manual intervention.

10. GPS Tracking

Capture GPS every 10 minutes during working hours and upload to CRM.

11. Attendance

Check-in/out with GPS, timestamp and optional photo.

12. Employee Dashboard

Show only assigned clients, pending work, follow-ups, payment and document status.

13. CRM Status Updates

Provide quick buttons for Documents Received, Client Not Responding, Payment Received, Waiting Government Approval, Completed and Callback Scheduled.

14-20. Security & Device Protection

Encrypt all data, block screenshots, detect root/debug/emulators, disable copy/share/export of client information, restrict browsers, Play Store, unknown APKs, critical settings and uninstallation using Android Enterprise Device Owner mode.

21. Factory Reset Protection

Implement Android Enterprise Device Owner with FRP. Normal APKs alone cannot enforce this.

22-24. Startup, Offline & Notifications

Auto-start after boot, encrypted offline storage with background sync, FCM push notifications.

25-27. Admin Dashboard & Device Control

Admins can monitor employee status, GPS, recordings, attendance, productivity and enforce device restrictions including USB, Bluetooth sharing, Nearby Share, file access and software installation.

28. Technology Stack

Kotlin, Jetpack Compose/XML, WorkManager, Encrypted Room DB, Retrofit, Android Keystore, REST APIs, JWT, HTTPS/TLS, RBAC, FCM, Google Location Services, Android Enterprise Device Owner Mode, Managed Google Play and MDM.

29. Workflow

CRM → Assign Client → APK Receives Task → Employee Opens Client → Click-to-Call → Call Recorded → Recording Uploaded → Status Updated → CRM Timeline Updated.

30. Important Technical Note

Requirements such as preventing factory reset, blocking app installation, disabling Settings and preventing uninstallation require Android Enterprise Device Owner mode with an MDM solution. These cannot be guaranteed by a standard APK.